The Journal of Open Source Software

Contextualized: Heterogeneous Modeling Toolbox

Caleb N. Ellington[1*¶], **Benjamin J. Lengerich**[2,3*¶], **Wesley Lo**[2,4], **Aaron Alvarez**[5], **Andrea Rubbi**[6], **Manolis Kellis**[2,3], **and Eric P. Xing**[1,7]

1 Carnegie Mellon University, USA **2** Massachusetts Institute of Technology, USA **3** Broad Institute of MIT and Harvard, USA **4** Worcester Polytechnic Institute, USA **5** University of Cincinnati, USA **6** Cambridge University, UK **7** Mohamed bin Zayed University of Artificial Intelligence, UAE **8** Petuum Inc., USA ¶ Corresponding author * These authors contributed equally.

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Fabian Scheipl

Reviewers:

- @holl-
- @pescap

Submitted: 12 February 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

Heterogeneous and context-dependent systems are common in real-world processes, such as those in biology, medicine, finance, and the social sciences. However, learning accurate and interpretable models of these heterogeneous systems remains an unsolved problem. Most statistical modeling approaches make strict assumptions about data homogeneity, leading to inaccurate models, while more flexible approaches are often too complex to interpret directly. Fundamentally, existing modeling tools force users to choose between accuracy and interpretability. Recent work on Contextualized Machine Learning (Lengerich et al., 2023) has introduced a new paradigm for modeling heterogeneous and context-dependent systems, which uses contextual metadata to generate sample-specific models, providing context-specific model-based insights and representing data heterogeneity with context-dependent model parameters.

Here, we present `Contextualized`, a SKLearn-style Python package for estimating and analyzing personalized context-dependent models based on Contextualized Machine Learning. `Contextualized` implements two reusable and extensible concepts: *a context encoder* which translates sample context or metadata into model parameters, and *sample-specific model* which is defined by the context-specific parameters. With the flexibility of context-dependent parameters, each context-specific model can be a simple model class, such as a linear or Gaussian model, providing direct model-based interpretability without sacrificing overall accuracy.

Statement of Need

"Personalized modeling" is a statistical method that has started to gain popularity in recent years for representing complex and heterogeneous systems exhibiting individual, sample-specific effects, such as those prevalent in complex diseases, financial markets, and social systems. In its basic form: $x_i \sim P(X_i; \theta_i)$, where i indexes a sample, θ_i is the parameters defining the sample-specific distribution, and x_i corresponds to the observation drawn from this sample-specific distribution, where understanding sample heterogeneity is equivalent to estimating data distributions with sample-specific parameters. Some methods, such as sample-left-out models (Kuijjer et al., 2019), provide sample-specific estimators without additional information but lack desirable statistical properties such as the ability to generalize to new samples or

test model performance on held-out data. Due to the difficulty of estimating sample-specific parameters, most methods make use of side information, covariates or "context," as an indicator of sample-to-sample variation (Fan & Zhang, 1999; Hastie & Tibshirani, 1993; Kolar et al., 2010; Parikh et al., 2011; Wang et al., 2022). However, prior methods only permit limited use of contextual side information, allowing models to vary over a few continuous covariates (Fan & Zhang, 1999; Hastie & Tibshirani, 1993; Wang et al., 2022), or a small number of groups (Kolar et al., 2010; Parikh et al., 2011; Zeileis et al., 2008), and do not scale to high-dimensional, complex, and sample-specific variation. `partykit` (Hothorn & Zeileis, 2015) goes beyond this by enabling random forests of model-based recursive partitioning (Zeileis et al., 2008) on contextual information, learning complex non-linear relationships between contextual information and linear regression parameters. However, `partykit` is limited to linear models and the relationship between linear coefficients and contextual information must be represented using tree-based methods, which struggle with high-dimensional and non-tabular data types. Recently, the contextual explanation network (CEN) was developed to learn this context-model relationship using a deep neural network, benefiting from a wide range of architectures targeting high-dimensional non-tabular data (Al-Shedivat et al., 2020). However, like model-based partitioning, the CEN is designed only for linear model personalization. Contextualized Machine Learning generalizes the CEN method, reframing the sample-specific parameter estimation problem as a more flexible and generalizable latent variable inference problem which provides a unified mathematical framework for inferring and estimating personalized models of heterogeneous and context-dependent systems using a wide range of model types (Lengerich et al., 2023).

Formally, Contexutalized Machine Learning uses subject data $X = \{X_i\}_{i=1}^N$ and context data $C = \{C_i\}_{i=1}^N$ where i indexes samples, we can express the likelihood of all data in the form of

$$P(X, C) \propto \int_\theta d\theta P_M(X \mid \theta) P(\theta \mid C)$$

where we call $P(\theta \mid C)$ the context encoder, and $P_M(X \mid \theta)$ the sample-specific model, where M denotes model class or type. So long as the choice for both the context encoder and sample-specific model are differentiable, we can learn to estimate parameters θ_i for each sample i via end-to-end backpropagation with gradient-based algorithms such that $P(X \mid C)$ is maximized. Conveniently, C can contain any multivariate or real features that are relevant to the study, such as clinical, genetic, textual, or image data, and the context encoder can be any differentiable function, such as a neural network, that maps C_i to θ_i.

`Contextualized` implements this framework for key types of context encoders and sample-specific models, opening up new avenues for quantitative analysis of complex and heterogeneous data, and simplifying the process of transforming this data into results with plug-and-play analysis tools. In particular, `Contextualized`:

1. **Unifies Modeling Frameworks:** `Contextualized` unifies modeling approaches for both homogeneous and heterogeneous data, including population models, varying-coefficient models (Fan & Zhang, 1999; Hastie & Tibshirani, 1993; Wang et al., 2022), and partition-based models (Kolar et al., 2010; Parikh et al., 2011; Zeileis et al., 2008) via context encoding, learning parameter variation over both continuous contexts and discrete groups. Additionally, `Contextualized` naturally falls back to these classic modeling frameworks when complex heterogeneity is not present. Not only is this convenient, but it limits the number of modeling decisions and validation tests required by users, reducing the risk of misspecification and false discoveries (Lengerich et al., 2023).
2. **Models High-resolution Heterogeneity:** Contextualized models adapt to the context of each sample by using a context encoder, naturally accounting for high-dimensional, continuous, and fine-grained variation between samples (Ellington et al., 2023).
3. **Quantifies Heterogeneity in Data:** Context-specific models quantify the randomness and structure of the systems underlying each data point, and variation in context-specific model parameters quantifies the heterogeneity between data points (Al-Shedivat et

al., 2018; Deuschel et al., 2023). `Contextualized` provides tools to analyze, test, and validate contextualized models, unlocking new studies of structured heterogeneity.

4. **Interpolates and Extrapolates to Unseen Contexts:** By using context encoders to translate between contextual information and model parameters, `Contextualized` learns meta-relationships between metadata and data. At test time, `Contextualized` can adapt to contexts which were never observed in the training data (Ellington et al., 2023).
5. **Analyzes Latent Processes:** By associating structured models with each sample, `Contextualized` enables analysis of samples with latent processes. These latent processes can be inferred from patterns in context-specific models, and can be used to identify latent subgroups, latent trajectories, and latent features that explain heterogeneity (Lengerich, Al-Shedivat, et al., 2022).
6. **Provides Direct Interpretability:** `Contextualized` estimates and analyzes context-specific statistical models. These statistical models are mathematically-constrained such that each parameter has specific meaning, permitting direct interpretation and immediate results (Lengerich, Nunnally, et al., 2022).
7. **Incorporates Multi-modal Data:** Context is a general and flexible concept, and context-encoders can be used to instill any type of contextual information into contextualized models, including images, text, tabular data, and more (Al-Shedivat et al., 2020; Lengerich et al., 2021; Lengerich, Al-Shedivat, et al., 2022; Stoica et al., 2020).
8. **Enables Modular Development:** The context encoder and sample-specific model within `Contextualized` are both highly adaptable; the context encoder can be replaced with any differentiable function, and any statistical model with a differentiable likelihood or log-likelihood can be contextualized and made sample-specific, benefiting from a rich ecosystem of statistical models and deep learning methods.

Usage

The `Contextualized` software is structured through three primary resources:

1. A simple plug-and-play interface to learn contextualized versions of popular model classes (e.g. classifiers, linear regression, graphical models, Gaussians).
2. A suite of context encoders to incorporate any modality of contextual data (e.g. continuous, categorical, images, text) and/or impose restrictions on context-dependent relationships (e.g. feature independence, interaction effects).
3. Intuitive analysis tools to understand, quantify, test, and visualize data with heterogeneous and context-dependent behavior. These tools focus on visualizing heterogeneity, automatic hypothesis testing, and feature selection for context-dependent and context-invariant features.

Installation instructions, tutorials, API reference, and open-source code are all available at contextualized.ml.

Acknowledgements

We are grateful for early user input from Juwayni Lucman, Alyssa Lee, and Jannik Deuschel.

References

Al-Shedivat, M., Dubey, A., & Xing, E. P. (2018). *Personalized Survival Prediction with Contextual Explanation Networks*. arXiv. https://doi.org/10.48550/arXiv.1801.09810

Al-Shedivat, M., Dubey, A., & Xing, E. P. (2020). *Contextual Explanation Networks*. arXiv. https://doi.org/10.48550/arXiv.1705.10301

Deuschel, J., Ellington, C. N., Lengerich, B. J., Luo, Y., Friederich, P., & Xing, E. P. (2023). *Contextualized Policy Recovery: Modeling and Interpreting Medical Decisions with Adaptive Imitation Learning*. arXiv. https://doi.org/10.48550/arXiv.2310.07918

Ellington, C. N., Lengerich, B. J., Watkins, T. B., Yang, J., Xiao, H., Kellis, M., & Xing, E. P. (2023). *Contextualized Networks Reveal Heterogeneous Transcriptomic Regulation in Tumors at Sample-Specific Resolution*. bioRxiv. https://doi.org/10.1101/2023.12.01.569658

Fan, J., & Zhang, W. (1999). Statistical estimation in varying coefficient models. *The Annals of Statistics*, *27*(5), 1491–1518. https://doi.org/10.1214/aos/1017939139

Hastie, T., & Tibshirani, R. (1993). Varying-Coefficient Models. *Journal of the Royal Statistical Society: Series B (Methodological)*, *55*(4), 757–779. https://doi.org/10.1111/j.2517-6161.1993.tb01939.x

Hothorn, T., & Zeileis, A. (2015). Partykit: A Modular Toolkit for Recursive Partytioning in R. *Journal of Machine Learning Research*, *16*(118), 3905–3909. http://jmlr.org/papers/v16/hothorn15a.html

Kolar, M., Song, L., Ahmed, A., & Xing, E. P. (2010). Estimating time-varying networks. *The Annals of Applied Statistics*, *4*(1). https://doi.org/10.1214/09-AOAS308

Kuijjer, M. L., Tung, M. G., Yuan, G., Quackenbush, J., & Glass, K. (2019). Estimating Sample-Specific Regulatory Networks. *iScience*, *14*, 226–240. https://doi.org/10.1016/j.isci.2019.03.021

Lengerich, B. J., Al-Shedivat, M., Alavi, A., Williams, J., Labbaki, S., & Xing, E. P. (2022). *Discriminative Subtyping of Lung Cancers from Histopathology Images via Contextual Deep Learning*. medRxiv. https://doi.org/10.1101/2020.06.25.20140053

Lengerich, B. J., Ellington, C. N., Aragam, B., Xing, E. P., & Kellis, M. (2021). *NOTMAD: Estimating Bayesian Networks with Sample-Specific Structures and Parameters*. arXiv. https://doi.org/10.48550/arXiv.2111.01104

Lengerich, B. J., Ellington, C. N., Rubbi, A., Kellis, M., & Xing, E. P. (2023). *Contextualized Machine Learning*. arXiv. https://doi.org/10.48550/arXiv.2310.11340

Lengerich, B. J., Nunnally, M. E., Aphinyanaphongs, Y., Ellington, C., & Caruana, R. (2022). Automated Interpretable Discovery of Heterogeneous Treatment Effectiveness: A COVID-19 Case Study. *J. Biomed. Inform.*, 104086. https://doi.org/10.1016/j.jbi.2022.104086

Parikh, A. P., Wu, W., Curtis, R. E., & Xing, E. P. (2011). TREEGL: Reverse engineering tree-evolving gene networks underlying developing biological lineages. *Bioinformatics*, *27*(13), i196–204. https://doi.org/10.1093/bioinformatics/btr239

Stoica, G., Stretcu, O., Platanios, E. A., Mitchell, T., & Póczos, B. (2020). Contextual Parameter Generation for Knowledge Graph Link Prediction. *Proceedings of the AAAI Conference on Artificial Intelligence*, *34*(03), 3000–3008. https://doi.org/10.1609/aaai.v34i03.5693

Wang, Z., Kaseb, A. O., Amin, H. M., Hassan, M. M., Wang, W., & Morris, J. S. (2022). Bayesian Edge Regression in Undirected Graphical Models to Characterize Interpatient Heterogeneity in Cancer. *Journal of the American Statistical Association*, *117*(538), 533–546. https://doi.org/10.1080/01621459.2021.2000866

Zeileis, A., Hothorn, T., & Hornik, K. (2008). Model-Based Recursive Partitioning. *Journal of Computational and Graphical Statistics*, *17*(2), 492–514. https://doi.org/10.1198/106186008X319331

DRAFT